

HAT-P-12b

R-1156

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_220308 · created 2026-07-31T01:48:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459646.8457 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.051
Transit depth (Rp/Rs)^2	2.67 +/- 1.68 [%]
Orbital Inclination (inc)	89.83 +/- 1.58
Airmass coefficient 1 (a1)	0.7144 +/- 0.0091
Airmass coefficient 2 (a2)	0.33 +/- 0.18
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	#2 - [337, 285]
Optimal Method	PSF photometry
Transit Duration (day)	0.0497 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.163 ± 0.051 vs 0.1406 (deviation 0.44σ)
Duration fit vs published	0.0497 d vs 0.0975 d (deviation 9.02σ)
Mid-time O-C	6.3 min
Depth SNR	3.2
Residual scatter	1.28 %

Light curve

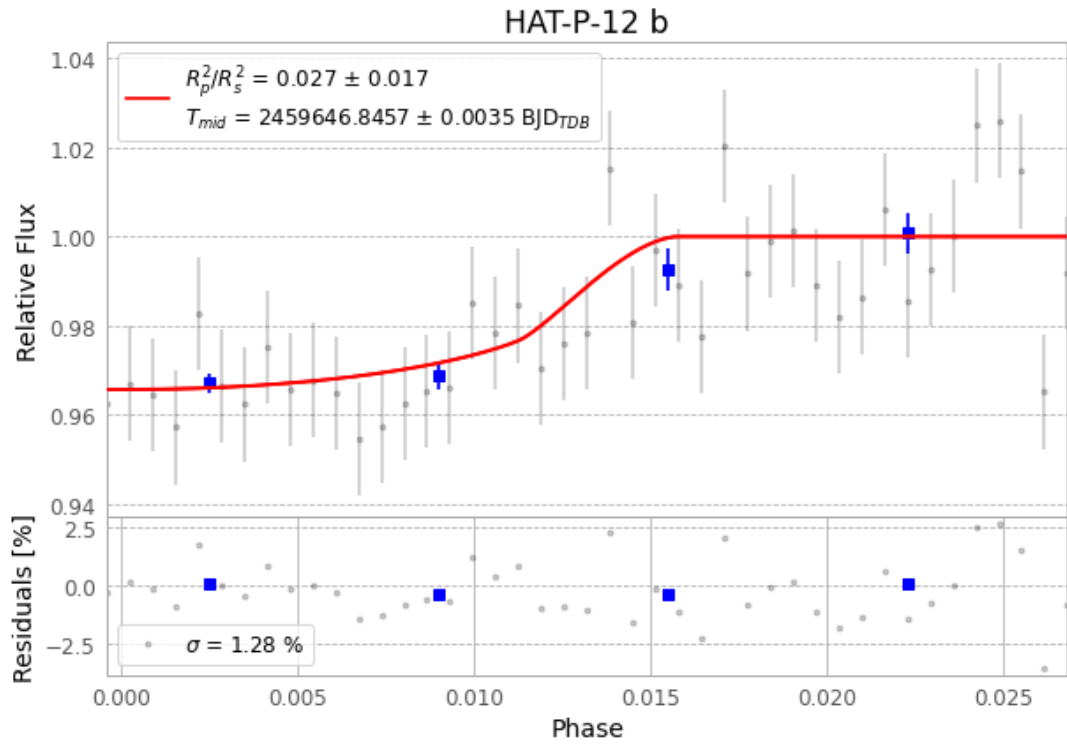


Figure 1 — FinalLightCurve_HAT-P-12 b_2022-03-08.png (SHA-256 566d7e88037e4df0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [213, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8907392dc17a7070...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

43 FITS frames (28 MB), HATP-12220308080912.FITS → HATP-12220308101513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-220308.log (7f9ee09b9356...), FinalLightCurve_HAT-P-12 b_2022-03-08.png (566d7e88037e...), FinalLightCurve_HAT-P-12 b_2022-03-08.csv (1910a69e7883...)

Compute resources

Wall time 90.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 01:48Z.